



AISHWARYA ARUNACHALAM

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AI Engineer

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Available Immediate (Remote)

Professional Summary

Results-driven Technology and AI professional with 11+ years of experience designing scalable systems and production-grade AI solutions. **Certified AI Engineer** specializing in generative AI, including RAG pipelines and agentic workflows that streamline complex operations and drive measurable efficiency gains. Proven ability to translate business requirements into high-impact, AI-enabled solutions through strong analytical thinking and cross-functional collaboration. Adept at delivering end-to-end automation frameworks that improve decision-making and support enterprise growth.

Core Competencies

- **Agentic AI and AI Agents:** Multi-agent orchestration (LangGraph, AutoGen), autonomous task execution, tool-use, MCP integration, supervisor-subagent design patterns.
- **Generative AI and LLMs:** RAG pipeline design, prompt engineering, fine-tuning (LoRA/RLHF), multi-modal AI, hallucination mitigation, structured output generation.
- **AI and LLM Engineering:** LangChain, LlamaIndex, Haystack, HuggingFace Transformers, AWS Bedrock, Google Vertex AI, Azure OpenAI, Claude API.
- **MLOps & Cloud Infrastructure:** Docker, Kubernetes, MLFlow, Airflow, GitHub Actions CI/CD; AWS, GCP, Azure; model monitoring, drift detection, and automated retraining pipelines.
- **Vector Databases & Retrieval:** Pinecone, ChromaDB, Weaviate, FAISS, pgvector; hybrid dense-sparse retrieval, re-ranking, and embedding pipeline optimisation.
- **Backend and Fullstack Development:** FastAPI, React, Node.js, REST/GraphQL, PostgreSQL, Redis; end-to-end product delivery from architecture to production deployment.

Projects

MediMitra: AI-Powered Dual-Role Healthcare Assistant

Tech: Mistral-7B-Instruct · pgvector · LangChain · S-PubMedBert-MS-MARCO · Surya-OCR · FastAPI · Streamlit · AWS RDS PostgreSQL

- Built an end-to-end FastAPI + Streamlit system combining a RAG symptom-to-doctor retrieval engine over ~900 biomedical vectors with privacy-first guardrails (no PII to external LLMs, hard refusal of diagnosis) and a Surya-OCR document intelligence pipeline for 4 healthcare/finance document types, formally evaluated and graded as a safety-compliant AI healthcare capstone at C-DAC (R&D MeitY, GoI).

Repo: github.com/aishwaryaarunachalam/MediMitra

Enterprise Knowledge Copilot with Governance

Tech: LangGraph · LangChain · pgvector · PostgreSQL · OpenAI/Anthropic/Groq LLMs · FastAPI · React · Keycloak/OAuth2 for RBAC

- Built a multi-agent RAG knowledge copilot over ~1k+ internal documents with policy-aware retrieval, access control, and observability, improving answer relevance and perceived quality by ~20–25% vs. baseline search in offline LLM-as-judge evaluations and pilot user tests.

Repo: github.com/aishwaryaarunachalam/EnterpriseKnowledgeCopilot

RAG & LLM Quality Lab (Evaluation and Testing Platform)

Tech: FastAPI · React · PostgreSQL · LangChain · RAGAS · LangSmith tracing · Docker

- Developed a self-hostable LLM evaluation and observability platform that runs 500+ test cases per app for faithfulness, hallucination, and safety, wiring results into CI/CD quality gates to block deployments when regression thresholds are breached.

Repo: github.com/aishwaryaarunachalam/LLMEvaluationLab

Multimodal Compliance & Content Intelligence Platform

Tech: Vision-Language Model (Gemini/GPT-4V class) · Surya-OCR · FastAPI · React · PostgreSQL · S3-compatible storage · Redis workers

- Implemented a multimodal compliance engine that analyzes PDFs, images, and short video clips for PII and policy violations, reducing manual moderation/review volume by an estimated 30–40% on pilot datasets while providing policy-aligned explanations and reviewer workflows.

Repo: github.com/aishwaryaarunachalam/MultimodalCompliancePlatform

Core Qualification

- Applied Machine Learning & Deep Learning:** Expert in designing, training, fine-tuning, and deploying supervised, unsupervised, and deep learning models (including Transformers).
- Data Engineering, Pipelines & Feature Work:** Advanced proficiency in building scalable data pipelines for ingestion, cleaning, transformation, and feature engineering using SQL/NoSQL, dataframes, and big-data frameworks.
- Model Evaluation, Experimentation & Monitoring:** Proven track record in defining robust metrics, running A/B tests and error analyses, and implementing monitoring for drift and degradation to improve model production performance.
- MLOps & Production AI Systems:** Highly skilled in operationalizing models with containerization, CI/CD, and scalable serving (Docker, Kubernetes, cloud AI platforms) to achieve reliable, low-latency, and cost-optimized inference at scale.
- LLMs, Generative AI & Prompt/Context Engineering:** Expert in integrating LLMs and other generative models with advanced prompt design, tool/agent orchestration, and guardrails to deliver accurate, safe, and efficient AI-powered user experiences.
- Retrieval, Knowledge & Context Systems (RAG):** Demonstrated success in architecting retrieval-augmented and knowledge-centric systems using vector search and hybrid retrieval to ground model outputs in authoritative organizational data.
- AI-First Product & System Design:** Proven ability to collaborate with product and UX to identify high-leverage AI use cases and design assistants, copilots, and recommendation flows that gracefully handle uncertainty and human override.
- Responsible & Secure AI Practice:** Deep understanding of privacy, security, and compliance, applying strict PII handling, access control, logging, and safety guardrails to ensure ethical and trustworthy AI systems.
- Cross-Functional Collaboration & Communication:** Demonstrated ability to translate ambiguous “AI ideas” into scoped, testable projects while clearly communicating model behavior, trade-offs, and risks to non-technical stakeholders.
- Continuous Learning & Adaptability in the AI Stack:** Highly proactive in tracking emerging models, frameworks, and infrastructure patterns and selectively upgrading the stack to maintain state-of-the-art solutions.

Achievements

- 1. Recognized as the Best-Performing Capstone Project in the CDAC cohort**
Recognized for: Technical implementation, innovation, and practical business impact
- 2. All India C-CAT Rank Holder - 223 - June 2025**
Recognized for: Securing All India Rank 223 in the highly competitive C-CAT examination.
- 3. Fiverr Ratings - 4.9 - Sep 2022**
Recognized for: Consistently delivering high-quality projects while maintaining a 4.9 client rating across engagements.
- 4. On the Spot Award - By Project Head at TCS - May 2015**
Recognized for: Exceptional documentation and going the extra mile for customer satisfaction.
- 5. Star of the Quarter - By Executive Director of Client Firm - Dec 2014**
Recognized for: Key contributions to a complex migration project and client query resolution.
- 6. On the Spot Award - By Group Manager at TCS - June 2014**
Recognized for: Rapidly mastering Sungard Intellimatch and delivering detail-oriented project work.

Client Collaboration Industries

FinTech & Payments · Healthcare & Life Sciences · Manufacturing & Industry 4.0 · Retail & E-Commerce · Banking & Credit Risk · Logistics & Supply Chain · EdTech & Learning Platforms · SaaS & Technology Products · Energy & Sustainability · Manpower & Recruitment · Legal & Compliance Tech · Government & Public Sector

Work Experience

- 1. Freelance Full stack Developer - **Self-Employed** - Bengaluru - **Sep 2021 – Apr 2025**
 - Demonstrated expertise in building and deploying scalable enterprise-grade applications using the MERN stack with AWS and Azure cloud implementations, delivering high-performance full stack solutions through microservices architecture, REST APIs, cloud-native deployments, CI/CD automation, database optimization, and secure infrastructure design for mission-critical business environments.
- 2. Freelance Digital Marketing Executive - **Self-Employed** - Bengaluru - **Jun 2016 – Jul 2021**
 - Proven ability to architect and optimize high-performance digital marketing ecosystems by managing multiple concurrent ad campaigns across channels, driving revenue growth and qualified lead generation through data-driven performance engineering, SEO optimization, conversion-focused automation workflows, nurture sequences, and continuous website performance improvements.
 - Worked as Technical Solutions Architect - **The Conversation Company** - Bengaluru - **Feb 2021 – Jul 2021**
- 3. Systems Engineer - **Tata Consultancy Services** - Chennai - **Nov 2013 – Feb 2016**
 - Proven track record in leading Agile-based reconciliation development and migration initiatives, successfully transforming a legacy in-house platform into Sungard Intellimatch for a US & UK investment banking client by delivering 4 major and 2 minor high-volume reconciliations processing millions of transactions with improved operational efficiency, accuracy and scalability.

Technical Stack & Tools

- LangChain
- TensorFlow, PyTorch
- Docker, Kubernetes
- AWS, Azure, GCP
- MLFlow, AirFlow
- Bedrock, Google Vertex
- Claude Code, CoPilot
- Libraries: Numpy, Pandas, Matplotlib, Seaborn, Keras, OpenCV, SpaCy
- KNIME, PowerBI

Education

- 2009-2013 Bachelor of Engineering (CSE) - Anna University, Chennai - **CGPA 8.39**
- 2009 HSC - **92.25%**
- 2007 SSLC - **91.5%**

Technical / Professional Certification / Microcredentials

- **Post Graduate Diploma in Artificial Intelligence - 900-hours fulltime**
CDAC - R&D MeitY (GoI) | **86.6%** | **A+ Grade** | **Aug 2025 - Feb 2026**
- Google Enterprise Agent Ready Certification (In Progress)
- Claude with Bedrock & Vertex AI, MCP, Agents, SubAgents, Claude Code, AI Frameworks - SkillJar by Anthropic
- KNIME AI Chatbots, RAG & Governance - KNIME
- Prompt Design in Vertex AI Skill, Google AI Essentials V1, Google Prompting Essentials - Coursera
- Digital Marketing Associate - Meta, Google, Amazon Ads, NSE's NCFM Foundation Module - Respective Direct Platforms

Personal Details

Present Address : Mangalore, India (Immediately Available Remotely)

Languages known : English, Tamil, Kannada

I hereby declare that all the details given above are true to the best of my knowledge and belief.